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Predicting policy funding allocation with Machine Learning

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ABSTRACT

Allocating funds through competitive opportunities is a core tool of place-based development policies, as it can generate economic benefits and support the revitalisation of 'left-behind' territories. By relying on Machine Learning (ML) techniques, this paper investigates the predictability of actors expected to benefit from EU development funding over the 2014–2020 period in Italy. We implemented eight different ML classification algorithms and Random Forest, followed by Extreme Gradient Boosting, and Support Vector Machine emerged as the most predictive. The results show that it is possible to make out-of-sample predictions and diagnose the precise factors influencing fund allocation, such as territorial attributes, economic dimensions, and production specialisation. Knowing in advance potential winners of the calls can help design tailored territorial, and even sectorial, public policies to address the obstacles to local development and green transition, and to efficiently distribute resources within the policy framework. This evidence contributes to the reflection launched by the Commission on the future of the competitiveness of the EU.

1. Introduction

Socioeconomic and environmental challenges have strengthened the case for sustainable territorial development to play a more central role within the European Union (EU), especially in rural and inner areas [1]. Governments have implemented several place-based and community-led interventions at both national and international levels to support sustainable development. Most of these rely on competitive funding opportunities, for which local actors, such as firms, are required to participate in calls for proposals. The EU's Resilience and Recovery Facility (RRF), in addition to the Cohesion Policy (CP) and the Rural Development Policy (RDP), have consensually emphasised the important role expected to be played by place-based interventions managed through competition procedures. For instance, a large portion of the funds from the *NextGenerationEU* is allocated through a direct management model, implemented by the European Commission (EC), by launching calls for proposals.[1]

The economic literature provides extensive evidence for the effect of place-based policies [2], albeit with some caution about the difficulty of achieving an efficient allocation of funds, especially in settings with structural and institutional weaknesses [3]. Nevertheless, in the literature, there is limited evidence related to the factors that may influence such allocations, and the effective interrelations and coordination among different funds, with most research in this area focusing on ex-post analyses of causal effects [4,5]. Even without examining the data, there are, in fact, good reasons to expect that some characteristics favour the likelihood of obtaining funds.

The general objective of this work is to explore the validity of using data to generate informational value for policy design. Specifically, the paper aims to implement a procedure for optimising policies based on competitive fund allocation by predicting the factors that determine the likelihood of being potential winners in such calls. It also demonstrates how this approach can be applied to a real EU place-sensitive policy for sustainable development that relies on competitive participation. Furthermore, the study provides insights into the interpretation of results, with a focus on how participants' heterogeneity can influence unintended policy effects.

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[1] There are several initiatives — economic and cultural activities — that are financed through the competitive allocation of funds throughout the EU at different territorial levels, from European and national to municipalities and city districts. For instance, the *Women TechEU* is a new initiative at the EU level to support women-led deep tech early-stage startups that are aimed at contributing to the green, digital, and social transition in line with the European objectives. Conversely, the *Estate Romana* is an example of a city-led call aiming to finance the implementation of cultural activities during the summer in Rome.

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As a case study of concrete EU policy, this paper looks at one of the main relevant structural funds through which the EU supports sustainable activities for socioeconomic development based on a competitive participation approach: the European Agricultural Fund for Rural Development (EAFRD) (Regulation EU 2021/2025). The analysis adopts machine learning (ML) models and is conducted at the micro-firm level and over the 2014–2020 period using Italy as case study.[2]

The results show that it is possible to make out-of-the-sample predictions regarding firms that are potential winners of competitive funding. Out of nine algorithms implemented, Random Forest, Extreme Gradient Boosting, and Support Vector Machine are the most powerful for this task. The most important features are related to the territorial characteristics, economic dimensions, and individual characteristics of the applicants.

This paper responds to the foregoing quest towards contributing to the present debate on the adoption of prediction models in policy design [6,7]. A competent anticipation and administration of funds allocation offers several unique benefits, particularly within the EU green and sustainable development strategies. Knowing in advance potential winners of the calls can provide policymakers with a valuable instrument to advance towards the policy objectives and efficiently distribute resources within the policy framework. In this context, a predictive model can offer additional market benefits by enabling strategic resource allocation, by improving the quality of proposals by applicants, and identifying potential competitors and collaborators. Furthermore, the capability of identifying promising beneficiaries with a high likelihood of securing EU funding can be attractive for investors and partners whilst also reducing information asymmetries.

Overall, this study offers a novel approach to policy evaluation by adopting ML to predict the allocation of EU development funds for the 2014–2020 period in Italy. By identifying key characteristics that influence fund allocation, this work enhances our understanding of resource distribution and provides a mechanism to evaluate the alignment of outcomes with the policy intention. A critical contribution is made through assessing whether the identified features reflect the policymaker's objectives. If these features align with the intended goals, it demonstrates that the policy tool is functioning as designed. In contrast, if unintended factors drive the outcomes, this highlights potential unintended impacts that warrant closer scrutiny. This dual perspective — evaluating both effectiveness and unintended consequences — positions the study as a valuable resource for researchers and policymakers. It offers a roadmap for refining place-based policy tools and adapting them to better serve the goals of sustainability and cohesion. By bridging methodological rigour with practical implications, the study contributes to advancing the integration of data-driven approaches in the design and evaluation of public policies.

The article is structured as follows. The next Section 2 situates the study in the literature and within the existing debate on place-based policies, with a focus on the policy framework under analysis. Section 3 presents the research design: data, sample, and methodology. Results are presented and discussed in Section 4 and Section 5, respectively, while Section 6 concludes the manuscript.

2. Literature review and contextual framework

Since this investigation takes the form of a multi-phase empirical study, the literature review served as a first step to clarify the contextual background of the analysis and to bring the research context into focus. Following the classification proposed by Paré et al. [8], in this paper, we conducted a narrative review, which attempted to identify what has been written on the subject. This type of review does not involve a systematic analysis of all the literature related to the topic, but rather only of the literature, and related research evidence (both from quantitative and qualitative methods), that are readily available. Results are, therefore, presented and synthesised discursively, with some type of commentary and interpretation by the authors. Even if a narrative review does not need to report on the paper searching and screening strategy which was adopted [8], our research started from designing a concise review protocol for completeness (see Appendix B).

2.1. Place-based policies

Place-based policies are expected to have the capacity to focus on the specific characteristics and needs of territories, as well as those of communities and actors living there, allowing for tailored solutions that can effectively address local needs [9]. A place-based approach involves considering two essential pillars. The first reflects the viewpoint that sustainable development should be a community-led undertaking. Without the engagement of socioeconomic actors, it would be impossible to implement an efficient transition pathway. The second position is that this process should be supported through context-sensitive measures, defined with reference to the geographical, historical, and socioeconomic features of the areas where they are implemented. In this sense, the rationale behind place-based policies is to account where and for whom they are needed, as diverse realities call for diverse types of interventions. The shifting from a *"one-to-fit-all"* towards a sensitive (*i.e.*, place and community based) approach to promote economic development entailed a long journey in the evolution of the European Union (EU) policy framework [10]. Presently, the concept that *"every place matter, and no one should be left behind"* is a flagship of the EU agenda, aiming at promoting a more inclusive, sustainable, and economically vibrant context in all the EU territories. Over the last few decades, in fact, this approach has been recognised as suitable for addressing not only economic, but also social and environmental considerations, promoting more sustainable practices and encouraging local participation [11,12]. Even if blind policies (*e.g.*, green infrastructure at national level) are necessary, they are, however, insufficient for addressing the structural change needed to convert local economic systems towards sustainability. To be effective, strategies should respond to specific needs while leveraging on unique strengths, and it is possible only by the adoption of a more tailored-made approach.

Despite the narrative suggesting that place-based policies enhance socioeconomic development, the empirical evidence is somewhat contradictory. A group of studies highlighted positive effects of place-based approaches on investments, employment, productivity, and wages [4,13–15]. In contrast, a second group of empirical analyses concludes, with a limited, or even negative, impact in some cases on local development, competitiveness, and total factor productivity [16–18].

What is more evident is that most of the existing studies — even if with different objectives and methodologies — have addressed this issue by looking at the ex-post evaluation of the effects of fund distribution [16–18]. On the factors that may influence the ex-ante fund allocation, there is, conversely, very little evidence, limited to investigating the role of only one, or, at most, a few, characteristics and conducted at the territorial level.

However, as reported by McCann [12]: *"today the underlying dynamics of modern interregional systems have changed profoundly [. . .] the conceptual and practical insights of place-based policies are needed more than ever"*. Therefore, it is reasonable to assume that these policies will continue to be adopted over an extended period. In this context, policymakers can use the ability to influence policy impacts ex-ante as a strategic tool by modifying the allocation and composition of EU funds. By favouring specific actors — based on ML predictions — policymakers can shape outcomes in a desired direction [6].

[2] This is the most recent programming period for which data on expenditures are available. For the current 2023–2027 CAP period, there are only data about the overall allocation at national level.

2.2. The place-based nature of the european agricultural fund for rural development

This study focuses on the European Agricultural Fund for Rural Development (EARDF), within the Rural Development Policy (RDP) framework, as a case of place-based policy reliant on competitive fund distribution among firms. This policy, as a case study, is an excellent choice for several reasons.

First, since the separation from the Cohesion Policy with the Agenda2000 reform, the EARDF has been the most important channel of the EU's local development support for 'left-behind' areas (Regulation EU 2021/2115). In the 2014–2020 programming period, the EAFRD budget was approximately €100 billion, which slightly decreased to €95.51 billion for the current 2023–2027 period. Since its inception, the EAFRD fund became the territorial pillar of the Common Agricultural Policy (CAP), the so-called Second Pillar, evolving from a sectorial towards a more place-based approach with a strong emphasis on rural development. However, despite its place-based nature, it remained linked to only one specific economic sector (*i.e.*, agrifood). This unique characteristic allows us to minimise the potential endogeneity, driven by the differences of applicants operating in different sectors, due to their exclusive responsibility for a specific sector [19].[3]

Second, with regard to investments in community-led and place-based interventions in left-behind areas, this has been reappraised as an efficient strategy for addressing sustainable development.[4] The correct management of these territories can be a strategic asset for reshaping the EU development trajectory. Contrasting food safety and security issues, the management of urban overpopulation, the generation of renewable energy, carbon storage, and habitats for wildlife can be listed as preliminary examples of the challenges to which rural areas could contribute.

Lastly, the EAFRD devoted specific attention to addressing climate change and environmental challenges, in line with the Commission's key priorities and strategies, such as the European Green Deal [20–22].[5] Economically weaker regions are more vulnerable to climate change mitigation strategies compared to their more prosperous counterparts and they, therefore, need public support to enable them to achieve the EU objectives [24]. Consequently, encouraging the sustainable management of natural resources and climate action has become one of the explicit objectives to which this fund contributes. For the 2023–2027 period, at least 30 percent of EAFRD must be dedicated to projects facing environmental issues [1]. This specificity enables us to effectually evaluate a place-based sustainability-oriented policy. Among the EAFRD measures, in fact, we select those that are exclusively targeting sustainable management of natural resources and climate actions.

This fund is implemented in shared management whereby both the European Commission (EC) and national authorities are in charge of running the programme. Financed projects are selected through specific calls for proposals, which define the total amount of allocated resources and the competitive process for their distribution among participants. National and regional authorities manage the selection of projects and the granting of payments. Local actors submit projects, which are then evaluated by Member States' administrations (at national, regional, and local levels) that select which projects to finance and authorise with responsibility for day-to-day management.[6]

Thus, the process is characterised by an insistence on structural redistribution among territories (in favour of rural areas) within which there is a competitive redistribution among actors (in favour of the most qualified projects and performing actors). Due to the spatial administration of these funds, the few papers that have debated the factors that may contribute to affecting how fund allocations are conducted at the territorial level. Among others, Pascucci et al. [25] focused on farmers' participation in rural development programmes, finding that it is affected by institutional priorities and geographic proximity. More recently, by aggregating beneficiary-level data at the NUTS3 level, Bonfiglio et al. [26] provided evidence on the relevance of economic capacity, while Camaioni et al. [27] focused on the effective capacity of the rurality degree and neighbouring space (the expenditure mix observed on a given unit is, in fact, dependent not only on the level of rurality, but also on the neighbouring units). Lastly, Vaquero-Piñeiro [28] looked at the role of territorial identity and demonstrated that the attractiveness of regional farming as a recipient of rural development funds was established on the funding body being responsive to territorial identity. The proximity of local producers, due to their being rooted in regional capabilities, helped to boost the inflow of rural development funds, both between regions and across sectors within regions.

However, even if it is difficult to consider all the potential determinants that could contribute to affecting fund allocation at the micro-level, and also the difficulty of obtaining such a vast amount of data for large samples, investigating economic phenomena, such as fund allocation, through the lens of firms is crucial [29].

3. Research design

3.1. Data and sample

This paper examines the characteristics of applicants who received funding for sustainable natural resource management and climate action through the EAFRD in Italy during the 2014–2020 programming period. The aim of the study is to disentangle the key factors influencing their likelihood of becoming potential winners. The analysis is based upon an *ad-hoc* firm-level database constructed by collecting data from the Italian Farm Accountancy Data Network (FADN). The FADN is a unique source of data at the micro-level; it is the only official EU collection of custom data based on harmonised bookkeeping principles for all EU countries.[7] Given the focus on the 2014–2020 programming period, the analysis relies on the 2014 wave that accounts for 7086 holdings, of which 47.67% received funds for climate action and sustainable management of resources (see Table 1 for the list of considered measures).

3.2. Methods

The dependent variable is defined as the probability of receiving at least €1 of EAFRD funding in the measures under analysis over the 2014–2020 period (1 if the unit obtains funds and zero other-

[3] Other place-based policies, such as Cohesion Funds, are not confined to specific sectors. Their territorial focus seeks to address challenges affecting a wide range of civil and economic stakeholders.

[4] Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee of the Regions. A long-term Vision for the EU's Rural Areas — is towards stronger, connected, resilient and prosperous rural areas by 2040. COM/2021/345.

[5] In the current 2023–2027 CAP strategic plans [23]: among the EAFRD initiatives, at least 30% of funding for each rural development programme must be dedicated to measures relevant for the environment and climate change [1].

[6] For instance, if a farmer or an agrifood firm have a project to start organic production, they would be eligible to apply for funds under the EARDF. For that, they would have to go through their country's institution, which is entrusted with managing the funds for projects on behalf of the EU and presenting the project to be selected.

[7] Extending the analysis to other EU countries is technically feasible, but data limitations are severe. Researchers have to formally request data from the government administrations responsible for managing the FADN database in each country. In Italy, the responsible administration is the *Consiglio per la ricerca in agricoltura e l'analisi dell'economia agraria* (CREA).

Table 1
Second Pillar CAP measures selected.

Measure	Definition
10.1	Agro-environment-climate commitments
11.1	Payment to convert to organic farming practices and methods
11.2	Payment to maintain organic farming practices and methods
12.1	Compensation payment for Natura2000 agricultural areas
13.1	Compensation payment for mountain areas
13.2	Compensation payment for other areas facing significant natural constraints
13.3	Compensation payment to other areas affected by specific constraints
15.1	Payment for forest-environmental and climate commitments
4.4	Non-productive investments linked to the achievement of agro-environment-climate objectives
8.1	Afforestation/creation of woodland
8.2	Establishment and maintenance of agro-forestry systems

Notes: in the analysis we considered both measures that target concrete actions and measures that refer to contextual conditions. Measures categorised as action are those that specifically target tangible steps towards fulfilling climate and environmental commitments. To receive funding under these measures, actors must actively 'do' something. Conversely, measures classified as conditions allocate funds based not only on the actors' actions ('what they do') but also on the structural and contextual conditions of their environment, which are outside the actors' direct control or decision-making. Action measures: 10.1, 11.1, 11.2, 15.1, 4.4, 8.1, 8.2. Condition measures: 12.1, 13.1, 13.2, 15.1.

wise).[8] Notwithstanding some limitation, these measures have been used in the literature as a good proxy for actions and environmental commitments [30].

Factor variables have been selected based on existing literature to reconstruct a representative sample of observable characteristics which could affect fund allocation from economic and territorial perspectives [31]. We identify 40 features that can be grouped in the following three clusters of factors potentially affecting the probability of receiving funds: (i) internal factors, (ii) economic activities, and (iii) territorial and political characteristics. The complete list is provided in Table A.1 in Appendix A, together with definitions. Variables are referred to the period before that under consideration [32].

From an empirical standpoint, we align with recent economic literature that argues by addressing policy issues, such as predicting which entities will benefit from funding opportunities; this requires minimising prediction error for which ML has proven to be particularly useful [6,33]. ML techniques are increasingly being used to address problems related to poverty targeting [34], evaluating the effectiveness of public programmes and spending [35], estimating Europeans firms' credit risk [36], and identifying corruption and political connections [37,38]. In the context of Italy, recent studies have harnessed the power of ML to predict the bankruptcy of local governments [39], vaccine hesitancy in municipalities [40], estimating local mortality and inequality during the Covid-19 pandemic [41,42], and predicting Geographical Indications areas [43]. Also, within the context of public policy design, ML may prove useful as shown by Battaglini et al. [44] and Battiston et al. [45]. In the former the authors evaluate individual's behaviour among self-employed and sole proprietorships in Italy showing how ML algorithms for classification may help in increasing tax revenues. The latter group of authors, instead, stress how ML improves the selection of taxpayers to audit. In both cases ML emerges as a strong tool to hinder tax evasion and, more generally, to assist policymakers for prediction-based policies. Even if these models use historical data, it can adapt to changes in policy design and allocation criteria as the features included in the model are not selected according to policy criteria, but rather they represent all the internal and external characteristics of firms. This means that a change in policy criteria and priorities does not affect the validity of the model.

The prediction task is formulated as follows: for each observation i (firm) at the time $t = 2014$, based on the set features $Features_{i,t}$ (see Table A.1 in the Appendix A for description and descriptive statistics), find the function $f(.)$ (ML model) that predicts the probability of receiving the funding $Fund_{i,T}$, where $T = [2014 : 2020]$ (Eq. (1)). This means that our features pertain to 2014, while the target variable — whether funding was received — covers the entire 2014–2020 period of EU development funding, therefore primarily the period following that of the features. We decided to rely on the entire time period 2014–2020 and not to use, for example, a panel data structure (*i.e.*, longitudinal yearly data for firms and funding allocation), because the funding allocation does not follow a strictly annual allocation: (i) there is no actual correspondence between the application for funds and the annual receipt of the same; (ii) the funds may refer to multiple applications; (iii) often the obtaining of the funds occurs all together at the end of the reference period.

$$\{Features_{i,t}\} \xrightarrow{f(.)} Fund_{i,T} \tag{1}$$

Following the predominant approach in the literature using ML models, we randomly divided the database as follows: 70% for training (train) and 30% for the out-of-sample testing set (test).[9] The hyper-parameter optimisation was only done on the training set using a repeated (10 times) ten-fold cross-validation. Moreover, we replicated this process five times by randomly performing the initial 30–70 split.[10] A similar approach, where the splitting procedure is reiterated several time, can be found in the work of Campedelli et al. [46]. To carry out our analysis, we used eight different ML predictive algorithms for supervised classification commonly implemented in literature [47–49]. In addition, for completeness, we also considered a more classical logistic regression (LR) as the baseline model. Within the ML classification task, the LR is implemented to test whether a specific task could be assessed by means of a relatively simple model without relying on complex algorithms. The ML algorithms implemented are presented below:

- Elastic Net (EN): a regression statistical method that performs features selection and regularisation with a mix of L1 (LASSO-type) and L2 (Ridge-type) penalisation to reduce over-fitting and increase prediction accuracy and interpretability. The LASSO (Least Absolute Shrinkage and Selection Operator) regression [50,51] is a shrinkage method which relies on regularisation to assess logistic issues when implemented for classification such as overfitting, multicollinearity, and feature selection. The penalty of the LASSO, differently from the Ridge regression, may reduce down to zero by preferring simpler models with fewer features. The Ridge regression [52], instead, is another regularisation algorithm which

[8] The €1 threshold is a standard criterion used in evaluating the effects of policy fund allocation. This is because any non-zero funding indicates participation in the funding programme. Specifically, for the fund under analysis, there is no minimum amount required to qualify; therefore, even receiving €1 means participation in the policy.

[9] Even though the split was causal, a sample coherence was established in the representatives of the farms among Italians regions.

[10] Results of the following tables and figures are obtained as the average of the outcomes from these five replications.

penalises the magnitude of coefficients while minimising the error between predicted and actual values. Nonetheless, in contrast to LASSO, the penalisation is never maximum, so no parameter is set to zero. In the grid search phase of our EN algorithm we tested the following parameters: `alpha`, the balancing between LASSO-L1 and Ridge-L2 regularisation and `lambda`, the regularisation's strength;

- Extreme gradient boosting (EGB): an advanced implementation of the gradient boosting ML technique [53], which is more efficient and scalable. It optimises the boosting process, using a more regularised model formalism and a parallelised algorithm for enhanced efficiency [54,55]. Gradient boosting (GB) algorithms fall within the wider umbrella of decision tree algorithms, non-parametric approaches developed to make decisions using a hierarchical, tree-based framework [56]. GBs are part of the boosting family, where the core idea is to fit weak learners sequentially. At each iteration, a new model is trained by accounting for the errors made by the previously fitted model with a penalisation to avoid overfitting. The 'extremisation' provided by EGB (also known as XGBoost) is due to the fact that while GB trains weak learners sequentially, EGB does it in parallel, resulting in a significantly faster computational speed. In the grid search phase of our EGB algorithm, we tested the following parameters: `nrounds`, the number of boosting iterations, hence trees, `eta`, the learning rate, `max_depth`, the maximum depth of the trees, `gamma`, the minimum loss reduction to make a further partition on a leaf node, `colsample_bytree`, the fraction of features sampled for each tree, `min_child_weight`, the minimum sum of instance weights in a child node, `subsample`, the fraction of observations sampled for each tree, `lambda`, the Ridge-L2 regularisation term on weights, which adds a penalty on large values of the model parameters to control overfitting, and `alpha`, the LASSO-L1 regularisation term on weights, which controls the model's sparsity;
- Flexible Discriminant Analysis (FDA): an extension of Linear Discriminant Analysis (LDA) that uses smooth functions to model non-linear decision boundaries between classes [57,58]. LDA, rooted in Fisher's discriminant analysis, finds a linear combination of features to best separate classes by assuming that the data follows a multivariate normal distribution with shared covariance across classes. Observations are classified based on Bayes' rule by calculating posterior probabilities. In contrast, FDA introduces greater flexibility by applying non-parametric techniques, such as smoothing functions, to model more complex, non-linear decision boundaries, which LDA cannot capture due to its linearity assumption. In the grid search phase of our FDA algorithm we tested the following parameters: `degree`, the degree of the polynomial used in the spline basis functions for non-linear feature transformation and `nprune`, the number of basis functions (or terms) to include in the model;
- k-Nearest Neighbours (KNN): a non-parametric algorithm that classifies a data point based on the majority class of its k-nearest neighbours in the feature space [59]. This algorithm operates under the principle that similar observations are likely to belong to the same class. To classify a data point, KNN first identifies the k closest data points in the training data using a specific distance metric, such as Euclidean or Manhattan distance. It then assigns a class label based on the most common label among the neighbours. For classification tasks, the label is assigned based on a majority vote, while for regression tasks, the average value of the neighbours is taken as the prediction. In the grid search phase of our KNN algorithm, we tested the following parameters: `kmax`, the maximum number of neighbours, `distance`, type of distance between data points, and `kernel`, the type of kernel function used to weigh the contributions of the neighbours;
- Naïve Bayes (NB): a probabilistic algorithm based on Bayes' theorem, assuming independence between features [60]. Despite the 'naïve' assumption that each feature independently contributes to the outcome, NB often performs well in practise, especially in high-dimensional spaces and natural language processing (NLP) tasks. The algorithm calculates the posterior probability of each class by combining the likelihood of observing the features given the class with the prior probability of the class. A Laplace correction (smoothing) factor is used to handle cases where certain feature values do not appear in the training set for a given class, thus avoiding zero-probability issues. NB is efficient, interpretable, and particularly suitable for tasks like text classification, where the independence assumption is often reasonable. During the grid search phase of our NB algorithm, we tested the following parameters: `adjust`, the bandwidth adjustment for kernel density estimation to avoid zero-probability issues, `usekernel`, which controls whether kernel density estimation is applied to continuous variables instead of assuming a normal distribution, and `fL`, which determines whether Laplace smoothing is applied;
- Neural Network (NN): a model that uses a set of connected input/output units in which each connection has a weight associated and learns by adjusting the weights to predict the correct class label of the given inputs [61–63]. NN algorithms are widely used in artificial intelligence applications which aims to emulate the behaviour of the human brain by using interconnected nodes or neurons in a layered structure. Input data is passed through an initial layer followed by several hidden layers with a feed-forward direction until achieving a final output layer which maps information from the last hidden layers into probabilities or values. Each neuron in a layer is characterised by a specific weight applied to the inputs as well as an activation function which decides whether a neuron should be activated or not. In the grid search phase of our NN algorithm we tested the following parameters: `decay`, a control for weight decay and `size`, the number of units, or neurons, in the hidden layer;
- Random Forest (RF): a family of randomised tree-based classifier decision trees that uses different random subsets of the features at each split in the tree [64]. RF is an ensemble learning method that combines multiple decision trees to improve predictive accuracy and control overfitting. Each tree in the forest is built using a randomly selected subset of both the training data (through bootstrap) and the input features. This randomness introduces diversity among the trees, ensuring that they capture different patterns in the data. Once all trees are trained, the final prediction is made by averaging the predictions in the case of regression, or through a majority vote for classification tasks. In the grid search phase of our RF algorithm, we tested the following parameters: `ntree`, the number of trees to fit and `mtry`, the number of features randomly selected in each split;
- Support Vector Machine (SVM): an algorithm that aims to find a hyperplane in a high-dimensional feature space that best separates data points while minimising the error between the predicted and actual target values [65,66]. The hyperplane serves as a decision boundary that classifies data points into different classes, and the optimal hyperplane identified by SVM maximises the margin between classes while minimising classification errors. The margin is defined as the distance between the hyperplane and the closest observations from each class, known as support vectors, because they define the decision boundary. SVM can effectively handle both linear and non-linear classification problems by employing specific kernel functions, which transform the data into a higher-dimensional space where it can become linearly separable. In the grid search phase of our SVM algorithm, we tested the following parameters: `sigma`, which controls the curvature of the decision boundary, or rather, the width of the radial basis function (RBF), and `C`, which is the penalty for misclassification.

The performance of our classification prediction — the probability of getting the funding — was assessed by analysing the Receiver Operating Characteristics curve (ROC) [67]. In our binary classification problem, the positive class (1) is defined as the firm which obtained the funding, and the negative class (0) as the firm which did not. The ROC curve serves as a visual representation of a classifier's performance across various discrimination thresholds by illustrating the trade-off between the true positive rate (Sensitivity), on the *y-axis*, and the false positive rate (1 - Specificity), on the *x-axis*.

Sensitivity (also known as Recall) represents the ratio between the number of firms predicted by the model as receiver of the fund which has effectively received it, known as true positives (TP), and the total number of firms which effectively received the fund, which comprises both the TP as well as those firms mistakenly classified by the model as not receiver of the fund, also known as false negatives (FN) (Eq. (2a)). This measure can be interpreted as the ability of a classifier to find all the positive units, or equivalently, to avoid false negatives. If we relate TP over the total number of firms predicted as receiver of funds, which comprised TP and the firms mistakenly classified by the model as receiver of the fund, also known as false positives (FP), we obtain another metric named Precision which is the ability of a classifier not to label as positive a unit that is negative (Eq. (2b)). Nonetheless, a limitation of both Sensitivity and Precision is that they only look at the relative size of the TP at the expense of the size of those firms correctly classified by the model as not receiver of funds, the true negatives (TN). For this purpose the Specificity index, instead, identifies the percentage of TN, over the total number of firms which effectively did not received the fund, which comprises both TN and FP (Eq. (2c)). Therefore, by considering the vertical axis of the ROC curve, 1 - Specificity represents the error to misclassify FP (Eq. (2d)) [49].

$$Sensitivity = \frac{TP}{TP + FN} \tag{2a}$$

$$Precision = \frac{TP}{TP + FP} \tag{2b}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{2c}$$

$$1 - Specificity = \frac{FP}{TN + FP} \tag{2d}$$

In ML, particularly in classification tasks, the ROC curve is a valuable tool for assessing the diagnostic capability of a model. As the classification threshold is adjusted, the ROC curve showcases how well the model distinguishes between positive and negative instances. In scenarios where the classification task is entirely unpredictable, the ROC curve aligns with the diagonal line, resulting in an Area Under the Curve (AUC) of 0.5. This value corresponds to classifying by random guessing. A model that performs perfectly would yield an AUC of 1, indicating flawless discrimination between classes. We can compare classifiers based on their AUC. In essence, a higher AUC signifies a more predictive model.[11]

4. Results

In this section, we present the results of the model that predicts those firms that are expected to benefit from EU funding. The focus will be on two main aspects: the predictability of our dependent variable (Section 4.1) and the importance of features of the independent variables used for predictions (Section 4.2).

4.1. Prediction performances

The ROC shown in Fig. 1 reveals that the RF model outperforms all the others. Each ROC is obtained as the average of the five ROCs (one for each time we performed the 30–70 split) obtained for each algorithm. Furthermore, the plotted curves are different from the 'classical' ROCs' representations since they are the result, for each algorithm, of five different tests. The graphical representation is obtained through a locally estimated scatterplot smoothing (LOESS) regression [68] over all ROCs values of each model. It is not possible, in fact, to simply represent the average values since, for each classification threshold level, specificity and sensitivity values change slightly among the models (obtained on different train sets).

The good predictability of RF is also confirmed by looking at other measures of performance, as shown in Table 2. In particular, the accuracy of RF is statistically higher than the no information rate. This is in line with previous empirical applications, confirming that tree-based models are among the more competitive methods for structured binary tasks [39,40,43]. The second-best performing model is EGB while the third place is taken by SVM. It is interesting to note that those algorithms are the best identified also by Berigel et al. [69] in their study to predict NEET (individuals between 15 and 29 age Not [engaged] in Education, Employment or Training) rates in Europe. Nonetheless, it should be stressed that, for all models, the accuracy is higher — and the difference is statistically significant — than for the no-information rate which represents the accuracy achievable by simply guessing the most frequent class.

Thereafter, once the satisfactory predictability of the models had been verified, we performed the DeLong test to determine whether the AUCs of the two models were statistically significantly different [71]. We compared the ROC curve of our best model (RF) with all other algorithms and results are shown in Table 3. Essentially, we want to test whether a specific model (RF) is better than another in terms of AUCs. Results of the DeLong test suggest not rejecting the null hypothesis at 1% for EGB and SVM. In other words, the performance differences among the top three algorithms (RF, EGB, and SVM) are not statistically significant. The evidence provided so far confirms the feasibility of using data to predict fund allocation among participants.

4.2. Features' importance

While in the previous section we identified the technical feasibility of predicting fund allocation according to firms' specificity, in this section we attempt to answer the following two questions: (i) which are the most relevant features in predicting fund allocation?; (ii) what is their relationship with the target variable?

To identify which are the most relevant features among those used for predictions, we classify them based on their importance, measured by the extent to which the model's performance decreased (*i.e.*, increase of the model's prediction error) when a specific feature was permuted during the prediction process. In doing so, we rely only on the best model (RF) and those models whose prediction performances were not statistically lower according to the DeLong's test (EGB and SVM as shown in Table 3).[12] The graphical representation of their relevance, depicted in Fig. 2, allows us to better present the core determinants in predicting fund allocation. All features are ordered by their relative importance, obtained as the average of the three algorithms considered. The two most important features refer to whether or not the potential beneficiaries are located on the plain (`plain`) or in mountain

[11] The analysis has been conducted in R using the package caret. For each selected algorithm (in parenthesis) we employed the following methods: `glm` (LR), `glmnet` (EN), `xgbTree` (EGB), `fda` (FDA), `kknn` (KNN), `naive_bayes` (NB), `pcaNNet` (NN), `rf` (RF), and `svmRadialSigma` (SVM).

[12] All features' importance has been first standardised and then ordered based on a weighted average constructed by multiplying each model's feature by the relative model's accuracy. The feature importance is calculated over the entire data frame (train and test) by selecting the best hyperparameters obtained from the trained models.

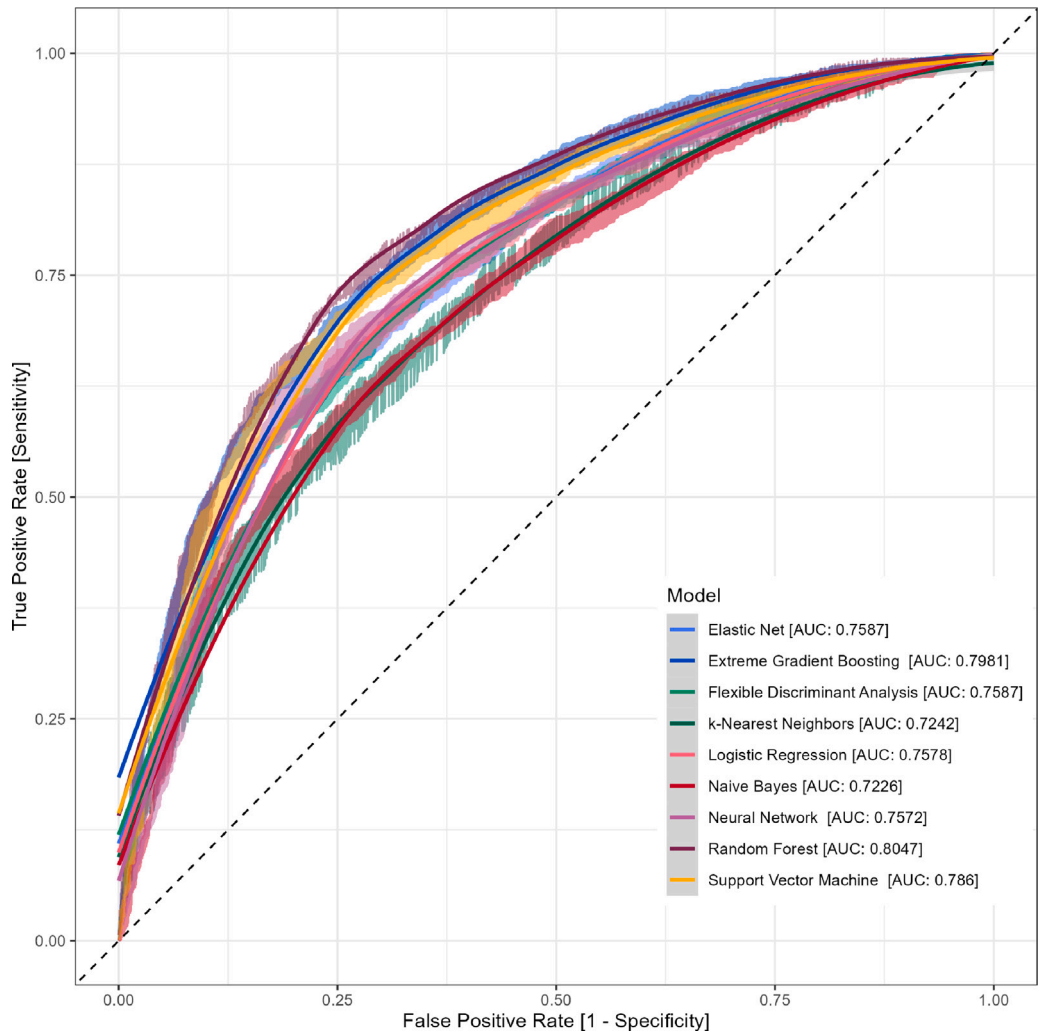

Fig. 1. Models' ROC curves.
Notes: models trained on 70% of observations and tested on the remaining 30%. The ROC curves were calculated on the test set (30% of the data). The curves were obtained as LOESS regressions of all ROCs values of each model.

Table 2
Models' performances.

Metrics	Logistic regression	Extreme gradient boosting	Flexible discriminant analysis	Elastic Net	k-Nearest neighbours	Naive bayes	Neural network	Random forest	Support vector machine
AUC	0.7578	0.7981	0.7587	0.7587	0.7242	0.7226	0.7572	0.8047	0.7860
Accuracy	0.6993	0.7267	0.6953	0.6975	0.6653	0.6259	0.7032	0.7376	0.7248
Kappa	0.3954	0.4506	0.3869	0.3916	0.3277	0.2313	0.4035	0.4729	0.4459
Accuracy Lower 95%	0.6793	0.7072	0.6752	0.6774	0.6447	0.6049	0.6832	0.7183	0.7052
Accuracy Upper 95%	0.7188	0.7456	0.7149	0.7170	0.6854	0.6465	0.7226	0.7562	0.7437
No Information Rate	0.5233	0.5233	0.5233	0.5233	0.5233	0.5233	0.5233	0.5233	0.5233
Accuracy *p*-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Mcnemar *p*-value	0.0248	0.0598	0.0101	0.0060	0.1866	0.0001	0.1624	0.1877	0.0001
Sensitivity	0.7440	0.7681	0.7494	0.7458	0.7024	0.8809	0.7401	0.7695	0.7833
Specificity	0.6502	0.6813	0.6360	0.6445	0.6247	0.3452	0.6624	0.7025	0.6605
Pos Pred Value	0.7002	0.7258	0.6934	0.6972	0.6727	0.6007	0.7067	0.7396	0.7170
Neg Pred Value	0.6982	0.7281	0.6981	0.6979	0.6565	0.7452	0.6991	0.7352	0.7352
Precision	0.7002	0.7258	0.6934	0.6972	0.6727	0.6007	0.7067	0.7396	0.7170
F1	0.7214	0.7462	0.7202	0.7206	0.6871	0.7105	0.7229	0.7542	0.7487
Prevalence	0.5233	0.5233	0.5233	0.5233	0.5233	0.5233	0.5233	0.5233	0.5233
Detection Rate	0.3893	0.4019	0.3922	0.3903	0.3675	0.4611	0.3873	0.4026	0.4099
Detection Prevalence	0.5561	0.5538	0.5657	0.5598	0.5464	0.7731	0.5482	0.5444	0.5718
Balanced Accuracy	0.6971	0.7247	0.6927	0.6951	0.6635	0.6131	0.7013	0.7360	0.7219

Notes: results are estimated on the confusion matrix, which shows a cross-tabulation of the observed and predicted classes, generating the predicted classes based on the typical 50% cutoff for the probabilities [70].

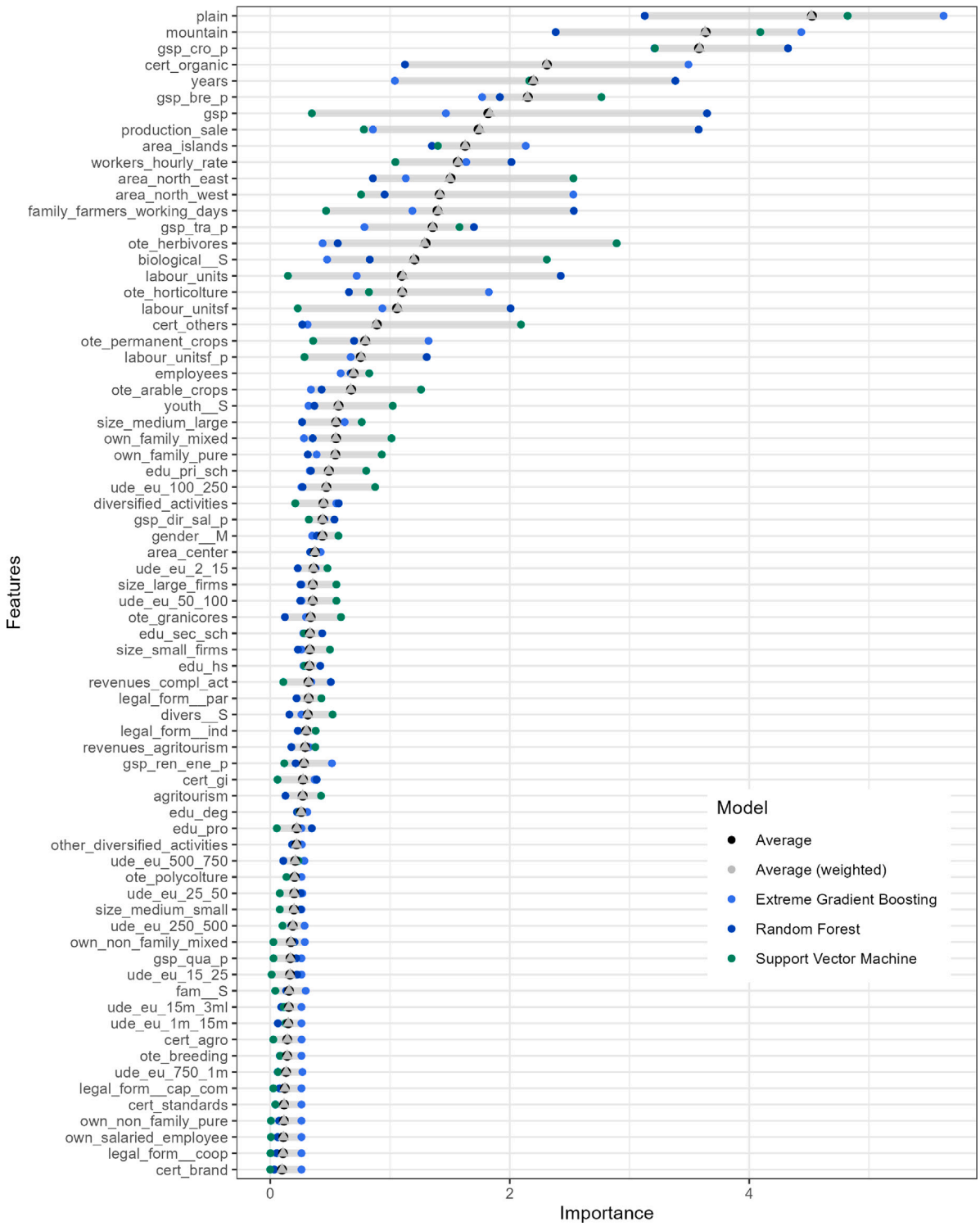

Fig. 2. Features' importance.

Notes: feature importance is computed across the entire dataset (train and test) using the best hyperparameters from the trained models. Importance values are rescaled across algorithms for comparability. For each algorithm, feature importance is calculated as the average across all data splits. The average (in black) values represent the simple mean of feature importance scores from the three algorithms. The average (weighted, in grey) values represent the mean feature importance scores weighted by the algorithms' (average) AUC values.

(`mountain`) areas followed by the share of gross salable production[13] in crops (`gsp_cro_p`). In fourth place is the possession of an organic production certificate `cert_organic` followed, in fifth place, by the entrepreneurs' ages (`years`).

[13] The GDP is expressed in economic value (€) and is linked to the main important revenue source for the farm. In the agricultural economic literature, this indicator is the main one relevant to accounting for the economic performance of farms.

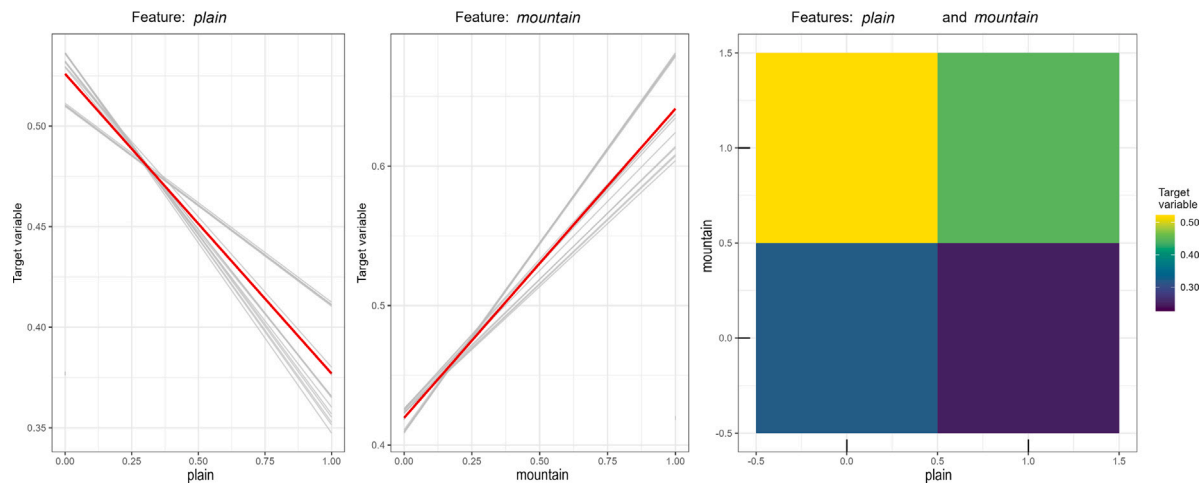

Fig. 3. Partial dependence plots for `plain` and `mountain`.
Notes: the PDPs are computed using the entire dataset (train and test). In the two tables on the left, the red lines represent the averages of the grey lines, which correspond to the PDPs calculated for each algorithm. These averages are weighted by the respective algorithm's AUC for each corresponding split.

Table 3
Models' DeLong test, RF against all other algorithms.

Model	stat z	*p*-value
Elastic Net	6.2272	0.0000
Extreme Gradient Boosting	1.3289	0.3206
Flexible Discriminant Analysis	6.3108	0.0000
k-Nearest Neighbours	8.6834	0.0000
Logistic Regression	6.2713	0.0000
Naive Bayes	8.9273	0.0000
Neural Network	6.4323	0.0000
Support Vector Machine	3.0794	0.0120

The directional relation of each feature with our target variable is, however, not captured by the feature importance. To provide some evidence in this respect, we need to look at the partial dependence plot (PDP) which graphically disentangles the functional relation between features and target variable. The PDPs are obtained by evaluating the model predictions while varying the selected feature over its range, holding all other features constant at their average [47].

In presenting the results—and then discussing them in the following Section 5—we focus exclusively on the category of factors that most significantly explain the success of call participation. The first group of core factors refers to territorial exogenous characteristics and to where potential beneficiaries are located in terms of altitude; the first element is being located in plain areas (`plain`), while the second is being located in mountain areas (`mountain`). The partial dependent plots (Fig. 3) show that applicants located in mountain areas are more likely to obtain funds, than those located in plain territories, who appeared to have less opportunity to attract funds

The second cluster of core determinants is related to the economic dimension and production specialisation of beneficiaries (`gsp_crop_p` and `gsp_breeding_p`). The economic value of saleable production emerges as a key driver, suggesting the relevance of the economic performance of participants in shaping the probability of obtaining funds. Even if they do not show a straightforward linear association, the overall impact follows a negative relation as shown in Fig. 4.

Applicants with higher economic size and a single main source of revenue obtain less funds, while the smaller that participants are and the more diversified the source of their revenues, the more likely they are to obtain funds for addressing environmental challenges.

This second group of characteristics also includes the choice to adopt more sustainability-oriented production processes, represented by the variable `cert_organic`, which reflects participation in the EU organic scheme[14] The PDP depicted by Fig. 5 shows a positive and increasing clear relation.

The final group of relevant factors is linked to the internal characteristics, such as the age and the education level of the holders, the employment structure and the business management type. Among them, the age of entrepreneurs (`years`) appears as the most relevant one and it shows an overall negative relation (Fig. 5): younger entrepreneurs are more likely to obtain funds.

5. Discussion

This paper is relevant for researchers interested in policy analysis for several reasons. The first finding of this study is the predictability of the dependent variable (*i.e.*, obtaining funds), as described in Section 4.1. In the case of place-based policies, it is possible to make out-of-sample predictions and thus predict which firms are more likely to benefit from EU funding based on their internal and external characteristics. This evidence should inform policymakers of the validity of conducting ex-ante predictive analysis through ML prediction models in policy targeting.

Second, the study shows the opportunity that using a ML approach offers for conducting, not only ex-ante projections, but also the ex-post validation of the policy. Thanks to this methodology, in fact, it is possible to compare the policy objective with the effective result in targeting specific actors. In this way, if the policy evidence (*i.e.*, who received funds) does not fit with the original aim, this suggests the presence of a potential unintended impact. In the future, therefore, a better targeting should be implemented in order to achieve policy targets, to mitigate potential bottlenecks, and to optimise the allocation of funds across different measures and policies. Conversely, if the policy aligns with the expected outcomes, this suggests that there will be no need to change it.

Third, the research contributes to the debate on the relevance of place-based policies by highlighting that funds allocation among participants is affected by their characteristics, including those that

[14] Organic certification refers, in fact, to more environmentally sustainable production processes implemented by farms: *"organic production is an overall system of farm management and food production that combines best environmental and climate action practices, a high level of biodiversity, the preservation of natural resources and the application of high animal welfare standards and high production standards"*. Regulation EU 2018/848 of the European Parliament and of the Council of May 30, 2018, on organic production and labelling of organic products and repealing Council Regulation No. 834/2007.

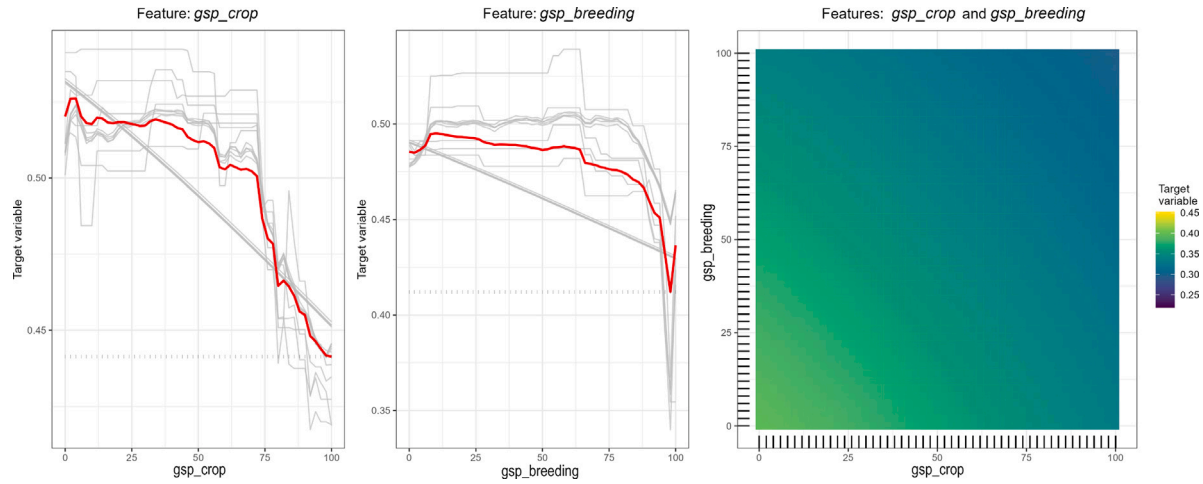

Fig. 4. Partial dependence plots for `gsp_crop_p` and `gsp_breeding_p`.
Notes: the PDPs are computed using the entire dataset (train and test). In the two tables on the left, the red lines represent the averages of the grey lines, which correspond to the PDPs calculated for each algorithm. These averages are weighted by the respective algorithm's AUC for each corresponding split.

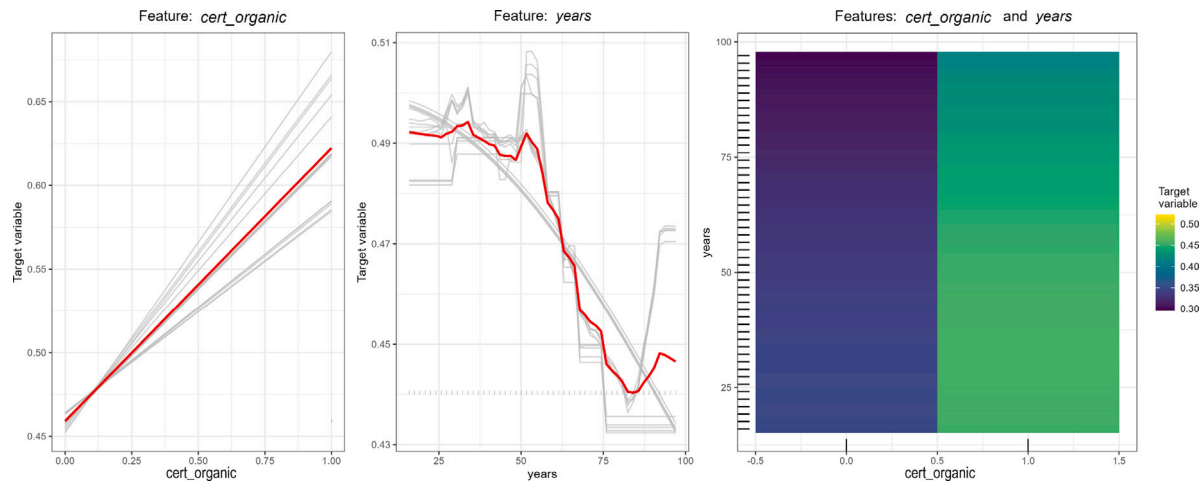

Fig. 5. Partial dependence plots for `cert_organic` and `years`.
Notes: the PDPs are computed using the entire dataset (train and test). In the two tables on the left, the red lines represent the averages of the grey lines, which correspond to the PDPs calculated for each algorithm. These averages are weighted by the respective algorithm's AUC for each corresponding split.

are not explicitly connected to the objectives of the call, which can be therefore positively or negatively related to the outcome.

In the case under analysis, the first evidence is that, for rural development projects, the geographical location of the main production site plays a significant role in its capacity to compete and attract funds (Fig. 3). Given the fund under analysis, these results can be linked to the increasing effort that the EU is making to support socioeconomic systems in mountain areas, which are, by definition, characterised by structural development difficulties [72]. How to foster the effective uptake of EU funds in mountain areas is a hot topic, not only for the rural development policies, but also, and even more so, for regional policies (*i.e.*, the EU Cohesion Policy).[15] Within a sustainable, cohesive and inclusive development framework, the EU aims, in fact, at supporting areas affected by population decline, lower added value economic activities and lower employment opportunities.

Contrastingly, the economic systems of land areas tend to be more market-oriented, to be open to adopting more intensive and mechanical-based production systems, and to benefit from structural facilities. This scenario seems to put applicants at a competitive disadvantage in terms of funds. However, this evidence invites reflections on two aspects. First, funds missing out in land areas could be a disadvantage especially for small and family-owned holdings, which tend to be less competitive in terms of sustainability performances if not supported. In contrast, larger farms can rely on economy of scale or private investments to achieve sustainable goals. Second, given their market-orientation strategies, economic actors located in land areas tend to be more focused on economic outcomes and competitiveness, with leading environmental sustainability not a priority. In this context, the lack of funds may exacerbate this trend at the expense of sustainable behaviours.

The negative relation between economic size, production specialisation and the opportunity to obtain funds (Fig. 4) suggests that funds can contribute to addressing the productivity gap between small and large firms. On the other hand, it reflects the EU perspective of considering economic systems diversification as a strategic channel through which supporting sustainable development and economic competitiveness at the local level, especially in Mediterranean countries, can be accomplished. In the light of our results, the difficulties faced by more specialised holdings in obtaining funds for environmental sustainability projects may be confirmed, therefore, in the future, arrogating public good provision (*e.g.*, environmental sustainability) to be predominantly linked to private initiatives and voluntary actions. In addition, spe-

[15] More information are available at: https://www.montana174.org/.

cialised farms are more exposed to sector-based vulnerabilities which can affect their economic and environmental resilience.

The relevance of participating in environmental voluntary schemes (Fig. 5) suggests that this commitment to sustainability generates benefits, not only for specific economic performances (*e.g.*, premium pricing) [73], but also in terms of fostering a strong competitiveness, which can positively affect funds' allocation. In the case under analysis, for example, producers that follow the EU organic regulation and decide to certify their productions are more inclined to successfully participate in other calls, even when such certification is not a mandatory requirement of the call. This encourages a reflection on the fact that investing in adopting more environmentally friendly production systems and communicating this choice through the EU certification schemes can generate a competitive advantage and external socioeconomic spill overs.

The determinant role played by the age of the firms' holders can be linked, at least in part, to the economic aspiration, active engagement and open-minded management approaches that characterised these actors. Generation renewal is one of the key objectives of both the EU CAP and Cohesion Policy, recognising their role in business and decision making, particularly in left-behind places. In addition, the literature has demonstrated that younger entrepreneurs are more sensitive about environmental issues [74]. Our result supports this need, showing that a generational renewal may improve, not only the performance of economic activities, as stressed by the literature, but also the opportunity of obtaining funds for resilience and sustainability.

While this study demonstrates the feasibility and effectiveness of using ML models to predict the allocation of competitive funding, it is important to consider the ethical implications of these approaches. One key concern lies in the potential for ML models to perpetuate or even exacerbate existing biases embedded in historical data. If not carefully addressed, these biases could lead to unfair outcomes, favouring certain types of firms or regions over others, irrespective of their current merit or potential. Moreover, preemptively determining likely winners of funding calls raises questions about the fairness and inclusivity of the allocation process. Policymakers must balance the efficiency gains offered by predictive analytics with the risk of reinforcing systemic inequalities. Drawing on recent work by Battiston et al. [45] on the optimisation of prediction-based policies, it becomes clear that transparency in model design, regular audits of outcomes, and the inclusion of fairness constraints in algorithms are crucial steps to mitigate these risks.

6. Conclusions

Although the adoption of place-based policies may not be sufficient to overcome the obstacles to economic development in left-behind and less developed areas, improving targeting remains a valid and effective approach. This study adopted a machine learning strategy to predict the likelihood for obtaining funds in competitive participation policies. In the analysis of the allocation of EAFRD funds in Italy, the evidence provided by this paper contributes to the academic debate as well as offering concrete recommendations for policy makers.

First, this paper demonstrated the validity of using a data-driven approach to create strategic informational value for policy design [75], welcoming policy makers to exploit these methodologies. The adoption of ML models seems to be a valid procedure to anticipate futures in policymaking, especially in respect of funds allocation. It is, in fact, possible to make out-of-sample predictions to identify firms that are likely to win competitive funding in the case of place-based policies.

In the case of territorial-sensitive interventions, there are several advantages to be gained by accurately predicting the winners in calls, from both political and sectorial perspectives. Thanks to this information, policymakers can have an anticipatory scenario of who will be financed by specific policies, based on the characteristics of those actors, and also to be able to indicate which are the weaknesses of those who are not expected to obtain funds.

Economic actors, like producers, can also benefit from the information that policymakers can achieve by adopting this methodology. They will be, in fact, more conscious of which specific factors work as catalysts for potential new beneficiaries to successfully compete for new funding opportunities. Furthermore, external investors can be more responsive by knowing the likelihood of not missing out on funds when they decide to set up a new activity or to invest in existing ones. In the long run, this more nuanced information can generate revenues that attract investments for left-behind areas where such funding has been lacking [76].

Today, the implementation of sensitive measures is more relevant than was ever the case previously, given the political debate on the future (post-2030) of the EU, at distinct levels. The EU is, in fact, still advocating in favour of instruments with a more markedly territorial vocation based on competition funding. As reported by the recent report *The future of European competitiveness* written by Mario Draghi in September 2024, two of the main challenges that the EU needs to face in the next years will be sustainability and competitiveness which could be addressed by implementing tailored measures and anticipating governments in public administration. In these decades of crisis, in fact, political decisions should be taken on the basis of evidence and in line with specific needs. Anticipating requires the implementation of tools that analyse, in depth, present and future trends. The approach proposed in this paper contributes to facing this challenge by identifying the use of data in future thinking practices [77]. It offers the opportunity of both conducting ex-ante predictive analysis of potential results and looking, ex-post, at the correspondence between policy targeting and its actual application. In this latter case, it can, therefore, serve as a strategic input in framing a policy issue highlighting where an objective is not being met, or where an emerging problem in fund allocation might be countered at an early stage Maffei et al. [77]. Even if foresight in government cannot define policy, it can help measures to be more appropriate and robust in their implementation and, therefore, the use of data to create informational knowledge for policy design will be more than welcome in the future Coates [78].

Second, by looking at a concrete case of local development policy, this study provided insights into the need of adopting place-based policies for local and rural development. According to our results, the distribution of funds is mediated based on participants' characteristics, even on those characteristics that are not directly targeted by the aim of the call. The key predictive features include territorial characteristics (such as location and regional attributes), economic dimensions (such as the financial and operational scale of the firms), and individual characteristics of the applicants (such as their qualifications and experience).

The debate on the post-2030 EU policies invite reflection on how the development of left-behind areas should be managed in the future, and on the contribution that rural areas can make to the green transition. In recent years, the washback from left-behind areas to the EU agenda has, in fact, been mainly linked to the renewed appreciation of the value of these areas in the fight against climate change and sustainability issues. Managing this role may require new governance mechanisms ensuring consistency across policies and a multi-level governance framework.

The idea that the Rural Development Policy (RDP) should not be considered in isolation, but rather consistently with other EU policies (*i.e.*, regional and environmental), is becoming a guiding principle in designing the road to the post 2030 era. The potential inefficiency effects of the cancellation of the interrelations between Cohesion and the Rural Development Fund [3] has, for example, reopened the debate about the future of the design of these policies. At the same time, the green transition should be supported by more ambitious environmental planning on the part of the Directorate-General for Climate Action (DG CLIMA) and for the Environment (DG ENV). The "greener" nature of the measures, and the related projects financed by the EAFRD, is,

in fact, not sufficient. Success will depend on the ability to establish an inclusive governance by involving all economic sectors, and local actors, in the design of strategic actions, which might converge in a completely new common policy.

In rethinking the development policy, the EU is proposing, therefore, a shift towards more centralised decision and funds management to face challenges in policies coordination, compliance and funds absorptions during the programming cycles. In this context, what clearly emerges from the analysis here presented, is that the EU institutions should design policies for local development without underestimating the effect of participant and territorial heterogeneity, which affect the probability of obtaining funds. Consequently, the assessment of the validity of a policy should consider, not only the efficiency of the policy, but also the social and territorial impact beyond the policy. In this context, an anticipatory government system approach, such as that proposed by this paper, can yield analytic benefits for policy design [79].

The main limitation of the research lies on the fact that data are not released concurrently with the development of the policy. This limits the promptness of the analysis and the opportunity of providing useful and timely insights to enable adjustments in the progress the policy. Our work only refers to a specific year regarding firm's characteristics and to one specific wave of EU funding (2014–2020). Therefore, it may be questioned whether it may be applicable in the future, also with a forecasting objective. Firms' characteristics change over time as well as the European agricultural sector within which this analysis was conducted; hence our model may be tailored for current policies but lacking in adaptivity for the future. Nonetheless, this limitation also represents a starting point for improvement for the future. By collecting firms and funding data on waves (or, where possible, yearly) basis, it would be possible to exploit the longitudinal information of data. It would require proper approaches to conducting out-of-sample validation without inflating the model by avoiding cross-section and /or temporal leakage as stressed by Cerqua et al. [80]. More generally, it would be ideal to develop a meta-learning model able to learn and improve on each new incoming data as proposed by Cerulli [81]. In this way, our model would be able to adapt to policy changes.

Despite the case study method proposed here, the same approach can be extended to other competitive measures of territorial-based development policies (*e.g.*, the Cohesion Policy) and to other EU countries. In addition, a heterogeneity analysis (*e.g.*, firms located in different areas or with different economic specialisations) could be useful to uncover factor-specific obstacles that may shape, even more effectively, the probability of obtaining funds. We leave these issues for future research.

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CRediT authorship contribution statement

Nicola Caravaggio: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Giuliano Resce:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Formal analysis, Conceptualization. **Cristina Vaquero-Piñeiro:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

See Table A.1.

Appendix B. Literature review protocol

In writing the literature review protocol used in this paper we followed Bandara and Syed [82].

Goal of the literature review

The aim of the paper is twofold:

- investigate the feasibility of using machine learning techniques to predict the allocation at the micro (firm) level through competitive processes of the EU funds for sustainable development projects (*i.e.*, sustainable management and climate actions);
- identify the most important features affecting the potential winners of competitive funding.

Key phenomena of interest

See Tables B.2 and B.3.

The search strategy

Where do you search in? Which fields will you include in your search? And which are the criteria adopted for initial search?

Regarding the sources in which we searched, we looked at online databases for academic articles (*e.g.*, Google Scholars, Scopus, Web of Science, Econopapers, Elicit, and others). We used a set of ad hoc keywords related to the key phenomena of interest, such as ML, Rural Development Policy, competitive funding, European Agricultural Fund for Rural Development, funds allocation.

In our research, we looked at the abstract, title or subject of the paper.

Regarding the common criteria, we refer to English-language papers being peer-reviewed published before August 2024. We do not set, in fact, time limits in the bibliographic search, due to the highly topical political debate of the topic under analysis and the scientific frontier of the methodological adopted.

In addition to peer-reviewed papers, we reviewed cross-references and cross-cited papers, national and international reports, working papers and conference contributions to control for potential publication impact [83,84].

Paper screening, tool and coding

Given that we conducted a narrative review, the review was not systematic and comprehensive and, therefore, the selection of the papers included in the sample was conducted through a selective approach.

Overall, in screening the papers, we decide to refer to:

1. studies discussing place-based policies, discarding those ones that do not introduce the potential socio-economic effects of this type of policies;

Table A.1
Descriptive statistics.

Variable name	Description	Mean	Median	Max	Min	sd
labour_units	Full-Time Equivalent	1.9734	1.4000	72.8200	0.0600	2.5389
workers_hourly_rate	Hourly rate	9.0880	8.0000	20.0000	0.0000	3.1518
diversified_activities	Other activities	4.3844	0.0000	112.0000	0.0000	15.6101
production_sale	Production sales (EUR)	1.2424	0.4351	187.5856	0.0000	3.9071
years	Years of activity	55.3950	55.0000	97.0000	16.0000	13.8295
agritourism	Farms engaged in agritourism activities	0.0512	0.0000	5.0000	0.0000	0.2782
other_diversified_activities	Farms engaged in diversified activities (different from agritourism)	0.0875	0.0000	4.0000	0.0000	0.3134
family_farmers_working_days	Working days (number)	408.0116	322.0000	2910.0000	10.0000	236.2578
employees	Number of workers (number)	1.8068	2.0000	10.0000	1.0000	0.9865
family_labour_units	Family workers (number)	1.3100	1.0000	9.0000	0.0200	0.7700
share_family_labour_units	Full-Time Equivalent for family workers (%)	83.9550	100.0000	100.0000	1.2658	25.9939
gsp	Gross Saleable Production (EUR)	1.4209	0.5357	207.3041	0.0000	4.2937
gsp_breeding	Gross Saleable Production from breeding (EUR) (%)	22.6921	0.0000	100.0000	0.0000	35.1399
gsp_crop	Gross Saleable Production from crops (EUR) (%)	63.4542	77.6730	100.0000	0.0000	35.4574
gsp_energy	Gross Saleable Production from renewable energy (EUR) (%)	0.6209	0.0000	100.0000	0.0000	6.5109
gsp_quality	Gross Saleable Production from quality products (EUR) (%)	2.2657	0.0000	100.0000	0.0000	13.3463
gsp_processed_food	Gross Saleable Production from processed products (EUR) (%)	9.2765	0.0000	100.0000	0.0000	21.9555
gsp_direct_sales	Gross Saleable Production from direct sales (EUR) (%)	3.0681	0.0000	100.0000	0.0000	13.7334
revenues_agritourism	Agritourism revenues (EUR) (%)	1.1629	0.0000	100.0000	0.0000	7.5861
revenues_compl_act	Complementary activities revenues (EUR) (%)	4.2080	0.0000	100.0000	0.0000	15.3275
cert_others	Other certifications	0.8428				
cert_agro	Agri-environmental certification(s)	0.0476				
cert_organic	Organic certification(s)	0.1169				
cert_standards	ISO-EMAS certification(s)	0.0059				
cert_brand	Private brand certification(s)	0.0006				
cert_gi	Geographical Indications	0.1955				
area_north_east	Macro area: North-East	0.2019				
area_north_west	Macro area: North-West	0.2457				
area_center	Macro area: Center	0.1559				
area_islands	Macro area: Islands	0.1181				
mountain	Area: Mountains	0.2319				
plain	Area: Hills	0.3268				
ote_herbivores	Companies specialising in herbivores	0.2303				
ote_permanent_crops	Companies specialising in permanent crops	0.3006				
ote_polycolture	Companies with polyculture	0.0642				
ote_arable_crops	Companies specialising in arable crops	0.2645				
ote_horticolture	Companies specialising in horticulture	0.0309				
ote_granicores	Companies specialising in granivores	0.0494				
ote_breeding	Companies with multi-breeding	0.0080				
size_small_firms	Small firm (economic size)	0.2303				
size_medium_small	Medium-small firm (economic size)	0.2178				
size_medium_large	Medium–large firm (economic size)	0.2583				
size_large_firms	Large firm (economic size)	0.2285				
ude_eu_2_15	Firm revenues: from 2 to <15 k€	0.1001				
ude_eu_15_25	Firm revenues: from 15 to <25 k€	0.1303				
ude_eu_25_50	Firm revenues: from 25 to <50 k€	0.2178				
ude_eu_50_100	Firm revenues: from 50 to <100 k€	0.2285				
ude_eu_100_250	Firm revenues: from 100 to <250 k€	0.1964				
ude_eu_250_500	Firm revenues: from 250 to <500 k€	0.0618				
ude_eu_500_750	Firm revenues: from 500 to <750 k€	0.0236				
ude_eu_750_1 m	Firm revenues: from 750 to <1 m€	0.0113				
ude_eu_1m_15 m	Firm revenues: from 1 m to <15 m€	0.0113				
ude_eu_15m_3 ml	Firm revenues: from 15 m to <30 m€	0.0106				
own_family_pure	Ownership: Family only	0.5048				
own_family_mixed	Ownership: Prevalence of family members	0.3477				
own_non_family_mixed	Ownership: Prevalence of extra-family members	0.1177				
own_non_family_pure	Ownership: Salaried workers	0.0155				
own_salaried_employee	Ownership: Contracting only	0.0064				
form_joint_stock	Legal form: Joint-stock company	0.0082				
form_partnership	Legal form: Partnership	0.1202				
form_individual	Legal form: Individual business	0.8666				
form_cooperative	Legal form: Cooperatives	0.0035				
owner_male	Owner gender: male	0.7892				
young	Young farmer	0.1461				
diversification_activities	Diversified activities (number)	0.1130				
organic	Organic farming	0.1167				
family_busines	Family business	0.9639				
edu_tertiary	Owner education level: Tertiary education	0.0567				
edu_high_shool	Owner education level: High school	0.2234				
edu_professional	Owner education level: Professional	0.1222				
edu_secondary	Owner education level: Secondary	0.3649				
edu_primary	Owner education level: Primary	0.1715				

Table B.2
Key phenomena of interest.

Key phenomena of interest	Synonyms and related concepts
Competitive funds allocation.	Call for tenders.
Machine Learning (ML) for predictions.	Out-of-the-sample predictions.
Place-based policies for sustainable development.	Territorial-sensitive, Community-led, Rural Development Policy, Cohesion Policy.

Table B.3
Research criteria.

Document type	Academic papers and official reports on: 1. place-based policies, especially studies debating on the effects of such policies on socio-economic development (provided evidence); 2. EU Rural Development Policy, especially studies debating on the factors that may contribute to affect fund allocation; 3. adoption of Machine Learning (ML) to address prediction policy problems, especially policy targeting and funds allocation.
Topic fields	Abstract, title or subject of the paper.
Criteria	Peer-reviewed, published, and unpublished papers.
Subject excluded	Studies on other place-based policies (e.g., Cohesion Policy) and competitive policies.
Time span	All years (up to August 2024).

2. studies on EU Rural Development and the related European Agricultural Fund for Rural Development, excluding papers generically.discussing rural development or rural development policies adopted in extra-EU countries;
3. studies on Machine Learning techniques, limiting to the power of ML for prediction. Regarding papers using machine learning, we considered studies discussing the adoption of ML to address prediction policy problems. We did not limit the sample at studies explicitly adopting ML for fund allocation, given that, to the best of our knowledge, there are only very few papers that addressed this issue.

To manage the data extraction and analysis process, we did not use a specific tool.

The coding of the papers has been conducted following the three macro themes: place-based policies, EU Rural Development Policies, and ML. In each of this group, papers have been distinguished between theoretical and empirical papers.

Data availability

Data will be made available on request.

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